EL NIÑO ACTION BRIEFING · 8 OCTOBER 2026

Elmwood, Berkeley

In Elmwood, El Niño's danger is a storm outage inside an old house: medical devices that stop, generators and grills brought indoors, and older people alone. The most important step is to reach the 102 residents who depend on powered medical equipment before December, with a battery that works in any outage.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
Alameda County, CA (inferred)	None published for the US, California, or the county	n/a here; global 6-month figure 239,000 ± 11,000	Storm power outages from trees on overhead lines; hillside landslides and a late fire window at the edges

11/11 recommendations checked. 9 corrected, 0 unverified, 2 contraindicated: a rainfall-triggered parametric policy, and invoking the state utility-shutoff protections now, when no qualifying emergency exists. None survived exactly as proposed. The funding placements were checked separately. One (CO alarms) is unverified and one (tree work) is confirmed as not to fund. **The funding target fell 26%**, from $201k to $148k, which is within the one-third tolerance. Exposure: Climate Impact Lab, 451,000 at Risk, September 2026, pre-peer-review. It publishes no figure for the United States, California, or Alameda

EXECUTIVE SUMMARY — AT A GLANCE



37.86°N 122.25°W, United States of America · Natural Earth

- **The risk.** The danger is a storm outage inside an old house: medical devices that stop, generators and grills brought indoors, and older people alone.
- **First action.** Berkeley OES, Berkeley Fire, and CERT reach the 102 residents on powered medical equipment by name, with a borrowable battery pool stocked by 30 November.*
- **The money.** $87k floor, $148k target, $260k ceiling; the floor buys 20 batteries at $1,500, Easy Does It $30k, Ashby Village $23k, and 100 alarms.
- **Do not.** Do not rely on the CPUC disaster utility protections being in force this winter: no trigger exists.

What to do	**7** actions	p. 2
Who does what	RACI chart	p. 9
Where money does the most	funding	p. 9
Do not do this	**2** items	p. 11
What has happened here	the record	p. 12
How this was produced	methodology	p. 14
References	sources	p. 15

* Suggested formulation: check with a health worker before sharing.

County, and heat is not this place's winter hazard. No population-scaled figure is given here.

Read this first: in Elmwood, a storm death will not be recorded as one. The verified record shows no Berkeley storm deaths in 2023. The people at risk here are 102 Medicare beneficiaries on electricity-dependent equipment (21 on oxygen) and 824 people aged 65 or over living alone, in a ZIP where 60% of homes were built before 1940. If they die, it will be from heart and lung causes on an outage day. After the 2003 New York blackout, the city attributed 6 deaths to it, while the excess-mortality analysis found about 90. So plan from a named list of people, not from a death rate. The verification also moved the hazard map. Most of the hillside slides the lenses cited (Middlefield, The Spiral, Wildcat Canyon) are in north Berkeley (94708), not Elmwood; only Alvarado Rd is in 94705. The Ashby Ave flooding in January 2023 was at 7th Street in West Berkeley. Elmwood's own flats are outside every CPUC High Fire-Threat District, which is exactly why PG&E's battery programmes do not reach them.

WHAT THIS PLACE SHOULD DO

RANKED

Actions are ranked by expected lives saved per unit of effort before the peak. The backing lenses are shown on each action, and the RACI chart below assigns each one. One rule underlies every action: trigger on individual atmospheric-river forecasts, not on the season. The strong 2015–16 El Niño left the East Bay below normal, and the record storms of 2022–23 came in a La Niña winter.

1. Reach the 102 residents on powered medical equipment by name, and stock a battery pool they can borrow in any outage, not only a wildfire shutoff

PARTIALLY CONFIRMED | PUBLIC HEALTH · EMERGENCY MGR · FRONTLINE · ECONOMIST · PHILANTHROPIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Berkeley OES with Berkeley Fire and CERT; PG&E for Medical Baseline; CIL or CRIL (whichever handles Berkeley) for the battery pool	Door-knocks and battery stock by 30 Nov	~$75k for 50 batteries (see funding); outreach is staff and volunteer time	Deaths from failed oxygen concentrators, ventilators, and other powered equipment during multi-day outages

* Suggested formulation: check with a health worker before sharing.

HHS emPOWER, queried directly on 8 October 2026, counts 3,516 Medicare beneficiaries in ZIP 94705. Of these, 102 use electricity-dependent equipment and 21 use oxygen. Fields reading “11” are suppression codes for counts of 1 to 10. The gap is a rules gap. PG&E's DDAR batteries are “only available during Public Safety Power Shutoffs” (fact sheet, 6 March 2026). The Portable Battery Program needs a shutoff or three or more EPSS outages since 2024, and it is closed for 2026 anyway. SGIP Equity Resiliency also shows closed. A CPUC map query puts College Ave and Ashby Ave outside any High Fire-Threat District, so a winter storm outage in the flats triggers none of these programmes. The action: knock on doors starting from the emPOWER count and the client lists of disability organisations, enrol people in Medical Baseline on the spot, write a move-out plan for outages longer than 24 hours, and fund a loan pool that works for any outage. PG&E Advice Letter 5566-E-A allows Medical Baseline contacts to be shared with local emergency agencies, but whether Berkeley OES has a data agreement is unverified. Do not wait for it.*

Corrections: the Center for Independent Living (CIL) is listed as the DDAR centre for Oakland and Alameda. Berkeley is assigned to CRIL in Hayward. Confirm which centre will run the pool. Transferability: the mortality evidence (Anderson & Bell, *Epidemiology* 2012: non-accidental deaths +25%, 95% CI 12–41%, about 90 excess deaths) comes from an August blackout in New York, where heat did much of the killing. The device-failure pathway transfers to a mild Berkeley winter; the heat effect does not. emPOWER counts Medicare beneficiaries only, so younger device users are missed, and 94705 also takes in part of Oakland.

2. Put the 824 people aged 65 or over who live alone on a storm-day check-in roster

PARTIALLY CONFIRMED | FRONTLINE · PUBLIC HEALTH · PHILANTHROPIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Ashby Village (phone check-ins), Elmwood block captains, Easy Does It for emergency attendants	**Roster live by 15 Nov**	**~$25k coordinator for 6 months; ~$40k Easy Does It storm surge**	**Undetected falls, dead devices, and cold homes during outages**

From the 2020–24 ACS for ZCTA 94705, re-pulled by the verifier: 3,723 residents are aged 65 or over (26.6%, against 16.4% county-wide), 1,449 are aged 75 or over, and 824 aged 65 or over live alone. Ashby Village (about 400 members, 300 volunteers, and $918k revenue in 2024–25) already runs screened volunteer phone check-ins for its members. The ask is to open that roster to non-members on storm days. Easy Does It (3222 Adeline St; $2.17M expenses in FY2025, funded mainly by Berkeley's Measure E) runs non-medical

* Suggested formulation: check with a health worker before sharing.

emergency attendants around the clock for disabled residents. A $40k grant is under 2% of its budget. Organisers note that the older resident alone in a large house, and the student in a converted basement, are on neither Nextdoor nor the neighbourhood association's mailing list. A neighbour's knock is often the only alert that reaches them.

Transferability: the evidence is extrapolated. Semenza et al. (*NEJM* 1996) found that living alone (OR 2.3) and not leaving home daily (OR 6.7) raised the risk of death in the 1995 Chicago heat wave, and that social contact protected. That is heat, not storms. For outages, the stronger argument is detection: someone notices the fall, the dead device, or the cold room. Ashby Village has not yet been asked whether it will open the roster to non-members.

3. Check and replace carbon-monoxide alarms, and tell renters and students in three languages: no generator, grill, or gas oven indoors

PARTIALLY CONFIRMED | PUBLIC HEALTH · SKEPTIC · PHILANTHROPIST · FRONTLINE | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Berkeley Fire with CERT and block volunteers; flyers in apartment lobbies on College and Ashby	**Before the first atmospheric river; flyers by 1 Nov**	**~$8k for 200 replacement alarms (unit cost unverified)**	**Carbon-monoxide deaths from fuel-burning heat and power sources used during outages**

California's SB 183 has required CO alarms in homes with fuel-burning appliances since 2011 (single-family) and 2013 (rentals), so landlords already owe the hardware. The gap is compliance, plus the alarms installed for those deadlines now reaching the end of their 7–10-year sensor life. After Hurricane Ike, generators accounted for 82–87% of CO exposures, and all six generator deaths had the unit inside the home or garage (CDC MMWR 2009). The local case: in January 2023, during a storm outage at Sea Ranch, two men died with a generator running in a hallway. Detectors could have prevented about half of the deaths among people found asleep in New Mexico (Yoon et al., *JAMA* 1998; this is a modelled estimate).*

* Suggested formulation: check with a health worker before sharing.

Corrections: the “more than 650 storm CO deaths, 2004–2012” figure was not found. CPSC counts 800 generator CO deaths over 1999–2012 and about 85 a year in 2011–2021, generator-related in general. The claim that “half of poisoned households had the generator outdoors” is not in the Ike report and has been dropped. The 581-outage figure in Worsham et al. (*NEJM* 2022) is behind a paywall and unverified. Call Berkeley Fire (510-981-2489) before buying alarms, as an existing programme may cover the hardware.

4. Survey the street trees over College, Ashby, and the Claremont Ave and Alvarado Rd routes, and push line clearances to PG&E before the first storm

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · FRONTLINE · SKEPTIC · ECONOMIST | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Parks, Recreation & Waterfront Urban Forestry Unit (city trees); PG&E (lines); volunteers reporting to 311 and trees@berkeleyca.gov	**Priority corridors by 1 Dec; a citywide survey is not realistic**	**Task orders on the City's as-needed tree contracts; ~$100–250k is a lens estimate, unsourced**	**The outages behind rank 1, and people trapped under trees and live lines**

The 21 March 2023 storm cut power to more than 15,000 Berkeley customers, and two people were trapped in a truck under a fallen tree and live line in the hills. Bay Area storm deaths that month included several people killed by trees falling on vehicles. The volunteer half of this action costs nothing and can start this week: walk the blocks, photograph leaning or cracked trees and limbs touching lines, report them, and keep a shared log so reports do not stall.

* Suggested formulation: check with a health worker before sharing.

Corrections: the 36-inch elm that fell on 3158 College Ave did so on 24 October 2016, not in 2023, and nobody was hurt. The owner is Urban Forestry in Parks, not Public Works. Measure FF (~$11.48M in FY2027) is fire, EMS, and wildfire-prevention money, not a street-tree fund, so treat it as a long shot limited to evacuation-route vegetation. No backlog figure is published; the unit's stated target is an arborist inspection within 2–4 weeks of a request.

5. Until the first soaking rain, enforce red-flag-day parking on the hill evacuation routes, reviving the stalled Safe Passages programme

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Berkeley Fire, with Police parking enforcement and tow contracts	Every Red Flag day from now until the first soaking rain	Staff time; no figure published	The 1991 failure: people dying on blocked roads out of the hills

If El Niño's rains arrive late, the Diablo-wind fire season runs into November and overlaps the start of storm season. The 1991 Tunnel fire (20 October; about 25 dead) reached Claremont Canyon at Elmwood's edge, and many of its dead were caught on blocked roads. Red-flag parking today is advisory. A council item in February 2025 asked for no-parking on fire-weather days with a report back by May 2025, and no adoption has been found; Berkeleyside reported in 2023 that Safe Passages had “largely languished.” Oakland runs a working version of this programme that can be copied.

* Suggested formulation: check with a health worker before sharing.

Corrections: the count of deaths on Charing Cross Road (9 or 11) could not be confirmed and has been dropped. The proposed EBMUD hydrant and generator drill was not verified and is left out. This is a fire action listed because of the timing overlap; it is not a storm action.

6. Give Alvarado Rd and the Claremont hills a local landslide trigger: NWS flash-flood watches with debris-flow wording, acted on through the existing evacuation zones

PARTIALLY CONFIRMED | HISTORIAN · EMERGENCY MGR | INVENTORY: WINTER WEATHER ADVISORIES (ANAND 2025), ADJACENT

OWNER	BY WHEN	COST	PREVENTS
Berkeley OES with Berkeley Fire; NWS San Francisco Bay Area for the watch	Trigger written by 1 Dec	Staff time	Fast shallow debris flows reaching homes and roads with no warning

The USGS and NWS ran a Bay Area debris-flow warning system from 1986 to 1995, issuing public warnings in February 1986 and watches in January 1993. Nothing has replaced it. The January 1982 storm set off more than 18,000 debris flows, and slides caused 25 of its 33 Bay Area deaths. Berkeley already has all-hazard evacuation zones (Genasys Protect) and 13 of 15 planned outdoor sirens installed. What is missing is a written rule saying which watch, plus how much rain, moves which zone.

Transferability: the slides the lenses cited are mostly in north Berkeley (94708), so this is an Elmwood-edge action, centred on Alvarado Rd. No landslide-specific zones exist, and no siren test schedule is published.

7. Re-clear storm-drain inlets on the old Derby and Potter creek lines before each forecast storm

PARTIALLY CONFIRMED | EMERGENCY MGR · HISTORIAN · SKEPTIC | INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
Public Works; residents through Adopt-A-Drain	Within 24 hours of each atmospheric-river forecast	Existing programme; the <$50k overtime figure is unverified	Street ponding where blocked inlets push water onto the road

The Potter and Derby watershed covers 3.8 square miles and runs almost entirely in culverts. Public Works already clears inlets year-round and sweeps leaves before the season, so the added value is targeted re-clears right before storms.

* Suggested formulation: check with a health worker before sharing.

Transferability: largely already in place, and no Elmwood culvert flooding is documented. Ranked last because it protects streets and property more than lives.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal.

Action	Berkeley OES	Berkeley Fire	Berkeley CERT	Police parking enforcement	Parks Urban Forestry	Public Works	PG&E	CIL / CRIL (DDAR)	Easy Does It	Ashby Village	Block captains & neighbourhood groups	Alameda County Public Health	NWS Bay Area
1. Named outreach to power-dependent residents; any-outage battery pool	**A**	**R**	**R**				**R**	**R**	**C**	**I**	**I**	**C**	
2. Storm-day check-in roster for people aged 65+ living alone	**I**		**C**						**R**	**A/R**	**R**		
3. CO alarm check-and-replace; generator, grill, and oven message	**C**	**A**	**R**				**C**				**R**	**C**	
4. Street-tree corridor survey; line clearances	**I**				**A/R**	**C**	**R**				**R**		
5. Red-flag-day parking on hill evacuation routes	**C**	**A**		**R**							**I**		**C**
6. Local landslide trigger for Alvarado Rd and the Claremont hills	**A**	**R**	**I**			**C**					**I**		**C**
7. Pre-storm inlet re-clears on the Derby and Potter lines						**A/R**					**R**		**I**

What the chart shows. **Berkeley OES is accountable for the action that matters most** (row 1), and that action depends on a Medical Baseline data agreement with PG&E that may not exist yet. **Block captains and neighbourhood groups carry an R on five of seven rows**, so whether the Elmwood associations are active (unconfirmed) is the plan's biggest dependency. **Which disability centre runs the battery pool is unresolved**: PG&E's directory assigns Berkeley to CRIL in Hayward, not to CIL on Adeline St.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR
$87k

TARGET
$148k

CEILING
$260k

LARGEST UNCERTAINTY
How many device users the existing PG&E programmes turn away, which sets the battery line

Floor: 20 batteries × $1,500 = $30k, plus Easy Does It $30k, Ashby Village $23k, and 100 alarms × $40 = $4k, for a total of $87k. **Target:** 50 × $1,500 = $75k, plus $40k, $25k, and 200 × $40 = $8k, for a total of $148k. **Ceiling:** 100 × $1,500 = $150k, plus $60k, $34k, and 400 × $40 =

* Suggested formulation: check with a health worker before sharing.

$16k, for a total of $260k. The target fell 26% in verification, from $201k, because 80 batteries against 102 device users was too high and new alarm installs duplicate a legal duty. The $1,500 battery and $40 alarm costs are plausible retail figures, not procurement quotes.

RECIPIENT	USE	AMOUNT	MECHANISM	BASIS	VERDICT
CIL or CRIL, whichever confirms it serves Berkeley	Action 1: battery loan pool for any outage	$75k	Pre-positioned stock	50 × $1,500 (a 1.5 kWh unit retails at $780–1,800); fills the gap that PG&E's shutoff-only rules leave	PARTIALLY CONFIRMED
Easy Does It Emergency Services	Action 2: storm-surge attendant and transport shifts, wheelchair battery swaps	$40k	Direct grant	1.8% of FY2025 expenses ($2.17M); no cost-per-life figure exists	PARTIALLY CONFIRMED
Ashby Village	Action 2: check-in roster opened to non-members	$25k	Direct grant	A ~$50k-a-year coordinator for 6 months. Willingness to open the roster is unconfirmed	PARTIALLY CONFIRMED
Berkeley Fire / CERT	Action 3: replace expired CO alarms for elders and renters	$8k	Pre-positioned stock	200 × $40, unsourced; landlords already owe the hardware under SB 183	UNVERIFIED
(none)	Action 4: tree work	$0	n/a	A City and PG&E duty; low and unquantified lives saved per philanthropic dollar	CONFIRMED: DO NOT FUND

The funder's one thing: most of this money does more good a mile away. emPOWER, queried in verification, counts 289 electricity-dependent Medicare users in 94702, 94703, and 94710 (South and West Berkeley), 2.8 times Elmwood's 102, at a higher rate (32–35 per 1,000 Medicare beneficiaries against 29). Easy Does It and CIL are both on Adeline St. Fund the battery and Easy Does It lines citywide, with at least half going south and west, and Elmwood's exposed minority is covered along the way. The lasting fix is regulatory: ask the CPUC to let DDAR batteries be lent in any outage, not only during wildfire shutoffs. A slower lever, Berkeley's seismic transfer-tax rebate (up to one-third of the base 1.5% transfer tax), could be extended by ordinance to retaining walls, drainage, and hazard trees. It cannot act before this winter's peak.

* Suggested formulation: check with a health worker before sharing.

DO NOT DO THIS CONTRAINDICATED

Advice that circulates for this hazard and would cost lives or waste the window here.

Buy parametric insurance triggered by 24-hour rainfall at the UC Berkeley gauge

CONTRAINDICATED AS WRITTEN

The Economist proposed a City-bought policy paying out to a neighbourhood tree-and-debris fund. The gauge exists (USC00040693, records from 1893), but NOAA returns no data after 31 August 2025, rows for 2023–25 have blank precipitation values, and it is read once a day by hand, so no rolling 24-hour total exists. One campus gauge is also a poor proxy for hillside failures, so the basis risk is high, and no comparable rainfall-indexed municipal product was found. It could not be priced and bound before December.

Instead: put this winter's money into rank 1. If the City wants risk transfer, scope it for 2027–28 on a maintained, real-time gauge network or a flood-extent index like Fremont's.

Rely on the CPUC disaster utility protections being in force this winter

CONTRAINDICATED: NO TRIGGER EXISTS

The protections under CPUC D.19-07-015 (no disconnections for non-payment, payment plans) need a Governor's or President's state of emergency *and* a disaster that disrupts or degrades utility service. A county declaration does not qualify. The 21 September 2026 proclamation the lenses cited *terminated* the 2023 storm emergencies in 13 counties, and Alameda is not among them. No state of emergency for the 2026–27 El Niño has been found.

Instead: treat it as contingent. Have Medical Baseline and CARE or FERA enrolment done in advance, and ask PG&E for an activation advice letter the day a qualifying state emergency covering Alameda County is declared. Golden State Water activated these protections on 23 December 2025.

Also avoid (lens advice, consistent with verified evidence): giving away or promoting generators, which drive storm CO poisonings; making sandbags the headline measure in flat Elmwood, which is not a floodwater area; and phone-only warnings, since the October 2019 shutoffs took 874 cell sites down. The generator point is backed by the verified Ike data; the other two were not separately verified.

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
17 Sep 1923	Berkeley fire, starting in Wildcat Canyon	~584 houses; no deaths	Stucco and tile replaced shingle; undersized water mains were exposed
3–5 Jan 1982	Bay Area storm: >18,000 debris flows across 10 counties	33 dead, 25 by landslide; no Berkeley deaths confirmed	USGS/NWS debris-flow warning system, 1986–1995, since discontinued
20 Oct 1991	Tunnel (Oakland Hills) fire, reaching Claremont Canyon at Elmwood's edge	~25 dead; ~2,843–3,000 homes	Hydrant incompatibility exposed; Safe Passages parking later proposed and stalled
1997–98	Strong El Niño	>$150M Bay Area landslide damage, >$20M in Alameda County (lens figures, not re-verified)	Mostly slow slides reactivating late in the season
2015–16	Strong El Niño that underdelivered	Oakland ~5.5 in by end of Feb against ~8 normal (lens figure)	Showed that a forecast El Niño is not a forecast storm
2016	36-inch city elm falls on 3158 College Ave	No injuries	None recorded
Dec 2022 – Jan 2023	Nine atmospheric rivers in a La Niña winter; 16 Jan slide at Middlefield (north Berkeley)	≥21 dead statewide; 5 homes red-tagged in Berkeley	None specific found
21–22 Mar 2023	“Bomb cyclone”	>15,000 Berkeley customers without power; two people trapped under a tree and live line	Berkeley asked PG&E again to underground evacuation-route lines; ~1 mile on Grizzly Peak funded

The recurrence nobody has noticed: the same hill slopes burn in dry autumns and slide in wet winters. Wildcat Canyon was the 1923 fire's origin, and Wildcat Canyon Rd was hit by the January 2023 slide almost exactly a century later. Burned or drought-stressed ground slides more

* Suggested formulation: check with a health worker before sharing.

readily, so a late-arriving El Niño after a dry autumn is the dangerous pairing. Berkeley still plans for fire and landslide separately, even though both depend on the same few narrow roads out.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Plan for El Niño** (most lenses) versus **El Niño barely shifts Berkeley's odds** (Skeptic). The 2015–16 event was dry, and the deadliest recent winter was a La Niña. The report adopts the Skeptic's fix rather than the Skeptic's conclusion: act on each atmospheric-river forecast, whatever the season label.
- **The danger is in the hills** (Historian, Emergency manager) versus **the danger is inside homes in the flats** (Public health, Frontline, Skeptic). Verification moved most of the cited slides to 94708 and found no recent Berkeley storm deaths. The flats' risk (devices, CO, isolation) is invisible in the death record by construction. Settling it needs an excess-mortality analysis of outage days, not a coroner's count.
- **Fund tree work** (Emergency manager, Economist) versus **don't fund it philanthropically** (Philanthropist). The verifier agreed it is a City and PG&E duty, so it is on the ranked list with a public owner and zero philanthropic dollars.
- **Spend in Elmwood** versus **spend in South and West Berkeley**. The emPOWER comparison supports the Philanthropist, so the funding section recommends citywide placements.

WHAT NOBODY COVERED — BLIND SPOT

No lens considered the **Hayward Fault**, which runs along the hills just east of Elmwood. An earthquake on saturated winter slopes is the compound event that would combine every hazard in this report: outages, slides, blocked roads, and fire. No lens covered **UC Berkeley students** in subdivided older houses either, beyond the flyer idea. The university's housing and alert channels were not examined. Smoke from a late-season fire elsewhere in the region was also left out.

HOW TO READ THIS — METHOD

The panel is author-built. Seven simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

* Suggested formulation: check with a health worker before sharing.

Mortality figures are resolved at Climate Impact Lab impact-region scale, roughly 300,000 people, and are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number.

HOW THIS BRIEFING WAS PRODUCED

METHODOLOGY

- **Scope.** The request named “Rocio, Berkeley”. No place of that name was found, and Angus confirmed Elmwood. Climate Impact Lab impact region, as this report records it: Alameda County, CA, *inferred* rather than resolved on the Impact Map. Most local statistics are for ZIP/ZCTA 94705, which covers Elmwood *plus* the Claremont hills and part of Oakland, about 14,000 people. Hill-specific findings are flagged where they belong to 94708 instead.
- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Heat is not this place's winter hazard, so the historical record carries the hazard instead.
- **Expert panel.** Seven simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, and philanthropist) each researched this place independently and in parallel, from web sources, to a fixed brief.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias before it counted. The lenses' converged “hills” narrative was demoted after verification placed most of its evidence in 94708. Contradictions were kept, including the DDAR battery contradiction, which the verifiers resolved with PG&E's own fact sheet and a CPUC map query. Actions were ranked by expected lives saved per unit of effort before the peak, given payers or marked unfunded, and reconciled against the funding placements.
- **Verification.** Five separate verifier agents checked 11 recommendations and the funding placements on evidence, transferability, and feasibility. Financial actions were also checked for availability, a named payer, a trigger, and whether they are already implemented. Verifiers queried HHS emPOWER and the CPUC High Fire-Threat District layer directly. The envelope was re-summed.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted, by driving the inventory's map interface in a browser earlier the same day (8 October 2026), because its download sits behind a sign-up form. The US filter returns 19 entries. The closest to any action here is *Winter weather advisories* (Anand

* Suggested formulation: check with a health worker before sharing.

2025: earlier advisories reduced vehicle crashes), cited for action 6. No US entry covers power-dependent residents, check-in rosters, CO alarms, tree work, or red-flag parking, so those actions are tagged “none found”.

- **Health wording.** Passages where a wording error could hurt someone (medical equipment in an outage, and generators and carbon monoxide) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.
- **Rules.** No cost is derived by dividing a published total. Retail and lens cost estimates are labelled as such. No local figure is invented; where data is thin, the report says so.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. No Elmwood-specific storm-death or outage record was found, so the outage evidence is Berkeley-wide. The 1997–98 and 2015–16 figures in the record come from the lenses and were not re-verified. Whether Elmwood's neighbourhood associations are active, whether Ashby Village will open its roster, and whether Berkeley OES receives Medical Baseline data are all unconfirmed. The EBMUD drill proposed by one lens was not verified and was dropped.

References · verified by Phase 5 unless marked. HHS emPOWER REST service (queried 8 Oct 2026) · Census Reporter, ZCTA 94705 (verified) · PG&E DDAR fact sheet (verified) · PG&E Portable Battery Program (verified) · SGIP programme metrics (verified) · PG&E Advice Letter 5566-E-A (verified) · Anderson & Bell 2012 (verified) · Semenza et al. 1996 (verified) · Easy Does It, IRS filings (verified) · Ashby Village, IRS filings (verified) · Ashby Village (verified) · CDC MMWR, Hurricane Ike CO (corrected) · Worsham et al., NEJM 2022 (exists; figures unverified) · Yoon et al., JAMA 1998 (verified) · CPSC generator CO report (verified) · Sea Ranch CO deaths (verified) · March 2023 storm, Berkeley Scanner (verified) · College Ave elm, KRON4 (corrected to 2016) · City trees (verified) · Measure FF (verified) · Ordinance 8,020-N.S. (verified) · Safe Passages, Feb 2025 (verified) · Safe Passages stalled, Berkeleyside (verified) · Oakland hills fire-safety parking (verified) · USGS Circular 1283 (verified) · USGS PP 1434, 1982 storm (lens, cited in verification) · Berkeley siren network (verified) · Jan 2023 Middlefield slide (verified) · Potter and Derby watersheds (verified) · Berkeley storm-season preparation (verified) · Jan 2023 outage and Ashby flooding (corrected) · Alta Bates future (verified) · UC Berkeley gauge record (verified) · Governor's termination proclamation, 21 Sep 2026 (verified) · Golden State Water activation precedent (verified) · Seismic transfer-tax rebate (verified) · CDI flood and mudslide coverage (verified) · Undergrounding requests (lens, not re-verified) · 1923 fire (lens, not re-verified) · USFA East Bay Hills fire report (verified in part) · El Niño rainfall years, CBS (lens, not re-verified)
Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory (US)
Generated 8 October 2026 for Elmwood, Berkeley. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.